Topics in Error-Correcting Codes

MIT 6.S953 — Problem Set 3

Due: November 18 at 6 PM.

Presentation instructions: Solve at least three of the problems. Only a presentation is required; no submission is required. Schedule your presentation here.

Throughout, $\Sigma = \mathbb{F}_q^s$, all dimensions are over $\mathbb{F}_q$. An additive code is an $\mathbb{F}_q$-linear subspace of Σ^n. For a linear space H, put $H_i = \{h \in H : h_i = 0\}$ and $H_S = \{h \in H : h_i = 0 \text{ for all } i \in S\}$.

Recall that C is a *τ-subspace-design code* if $\tau : \mathbb{N} \to [0,1]$ is nondecreasing and, for every $r \geq 1$ and every linear $H \subseteq C$ with $\dim H \leq r$,

$$\frac{1}{n}\sum_{i=1}^{n} \dim H_i \leq \tau(r) \dim H.$$

Problem 1: The folded Wronskian and a sharper subspace-design bound

Let $1 \leq k \leq sn$, and let $\gamma \in \mathbb{F}_q^*$ have order at least k. Suppose all sn points $\alpha_i \gamma^a$, for $i \in [n]$ and $0 \leq a < s$, are distinct and nonzero. The folded Reed–Solomon code is

$$C = \left\{ \big(f(\alpha_i), f(\alpha_i\gamma), \ldots, f(\alpha_i\gamma^{s-1})\big)_{i=1}^{n} : f \in \mathbb{F}_q[X],\ \deg f < k \right\}.$$

Its rate is $R = k/(sn)$. For $1 \leq r \leq \min\{k, s\}$ and $f_1, \ldots, f_r$ of degree less than k, define

$$W_f(X) = \det\big(f_j(\gamma^a X)\big)_{0 \leq a < r,\ 1 \leq j \leq r}.$$

(a) **The Wronskian criterion.** Prove that $W_f \neq 0$ if and only if $f_1, \ldots, f_r$ are linearly independent over $\mathbb{F}_q$.

Hint: Reduce to working with $f_1, \ldots, f_r$ of distinct degrees $d_1 < \cdots < d_r$ and show that the coefficient of $X^{\sum_j d_j}$ is non-zero. Why do we need γ to have large multiplicative order?

(b) **Highest and lowest exponents.** For independent $f_1, \ldots, f_r$, prove

$$\deg W_f \leq kr - \binom{r+1}{2}, \qquad X^{\binom{r}{2}} \mid W_f.$$

Deduce that the number of nonzero roots, counted with multiplicity, is at most $r(k-r)$.

(c) **Root multiplicities from a block.** Let $H \subseteq C$ have dimension r, identify it with its message polynomial space, and form the Wronskian of a basis. For $r_i = \dim H_i$, show that

$$(X - \alpha_i\gamma^u)^{r_i} \mid W_f(X) \qquad (0 \leq u \leq s - r).$$

(d) **The sharper parameter.** Deduce $\sum_i \dim H_i \leq r(k-r)/(s-r+1)$. If $k \geq s+1$, conclude that one may take

$$\tau(r) = \min\left\{1, \frac{k-r}{n(s-r+1)}\right\} \quad (1 \leq r \leq s), \qquad \tau(r) = 1 \quad (r > s).$$

Problem 2: Pruning and list decoding

(a) **Uniform sampling: a distance-only bound.** Let $H \subseteq \Sigma^n$ be a linear space with dimension r and relative minimum distance at least δ. For $0 < \varepsilon \le \delta$ and every $y \in \Sigma^n$, prove

$$|\mathsf{B}_{\delta-\varepsilon}(y) \cap H| \le \varepsilon^{-r}.$$

Give a randomized procedure outputting at most one element of H that returns each fixed $c \in \mathsf{B}_{\delta-\varepsilon}(y) \cap H$ with probability at least ε^r.

Hint: Sample a uniform coordinate and impose $x_i = y_i$, and repeat this procedure for r steps.

(b) **The design property improves the list bound.** Let C be a τ-subspace-design code and $\mathcal{A} \subseteq C$ an affine space of dimension r. Fix $0 < \varepsilon < 1 - \tau(r)$ and put $\rho = 1 - \tau(r) - \varepsilon$. Prove

$$|\mathsf{B}_\rho(y) \cap \mathcal{A}| \le \frac{r+\varepsilon}{\varepsilon}.$$

Hint: Induct on the current dimension $d \le r$ and double count number of agreements.

(c) **Algorithmizing the induction.** Start with $\mathcal{A}' = \mathcal{A}$. If $\dim \mathcal{A}' = 0$, return its unique point. At a nonempty affine space $\mathcal{A}'$ of dimension $d > 0$, let

$$A_i = \{x \in \mathcal{A}' : x_i = y_i\}, \qquad w_i = \begin{cases} \dim A_i + \varepsilon, & A_i \neq \varnothing \text{ and } \dim A_i < d, \\ 0, & \text{otherwise.} \end{cases}$$

If $\sum_j w_j = 0$, return failure. Otherwise, sample i with probability $w_i / \sum_j w_j$ and recurse on A_i. Show that the procedure takes at most r steps and returns each $c \in \mathsf{B}_\rho(y) \cap \mathcal{A}$ with probability at least $\varepsilon/(r+\varepsilon)$.

(d) Show that for a τ-subspace design code C, the code is $(1 - \tau(\lceil 1/\varepsilon \rceil) - \varepsilon, O(1/\varepsilon^2))$ list-decodable.

Problem 3: Containment above the LCL threshold

Let $(V_1, \ldots, V_n)$ be an r-LCL profile, where each $V_i : \mathbb{F}_q^r \to \mathbb{F}_q^r$ is linear. Recall that

$$\Phi_W((V_1, \ldots, V_n), \alpha) = \alpha \dim W - \frac{1}{n} \sum_{i=1}^{n} \big(\dim W - \dim(W \cap \ker V_i)\big).$$

Recall that C contains the profile if there are linearly independent codewords $c_1, \ldots, c_r \in C$ such that $V_i(c_{1,i}, \ldots, c_{r,i}) = 0$ for all $i \in [n]$. Suppose that for every proper subspace $W \subsetneq \mathbb{F}_q^r$, $\Phi_{\mathbb{F}_q^r}((V_1, \ldots, V_n), R_*) \ge \Phi_W((V_1, \ldots, V_n), R_*)$.

Let $C \subseteq \mathbb{F}_q^n$ be a rate $R = R_* + \varepsilon$ random linear-code. Prove that

$$\Pr[\mathcal{C} \text{ contains } (V_1, \ldots, V_n)] \ge 1 - q^{r^2 - n\varepsilon}.$$

Write $\Phi_W = \Phi_W((V_1, \ldots, V_n), R)$.

For $W \subseteq \mathbb{F}_q^r$, let X_W count the matrices $M = \begin{bmatrix} c_1 \\ \vdots \\ c_r \end{bmatrix}$ with codewords $c_1, \ldots, c_r \in C$, satisfying:

- $V_i(c_{1,i}, \ldots, c_{r,i}) = 0 \quad (i \in [n])$.
- $\mathsf{span}_{\mathbb{F}_q}((c_{1,i}, \ldots, c_{r,i})_{i \in [n]}) = W$.

(a) **An expectation bound.** Show that

$$\mathbb{E}[X_W] \leq q^{n\Phi_W}.$$

(b) **A deterministic lower bound.** Show that for every rate R linear-code

$$\sum_{W \leq \mathbb{F}_q^r} X_W \geq q^{n\Phi_{\mathbb{F}_q^r}}.$$

(c) **Full-rank containment.** Combine parts (a) and (b) with Markov's inequality to prove the claimed bound.

Problem 4: Entropic Brascamp-Lieb and zero-error list recovery

A code $C \subseteq \Sigma^n$ is said to be (ℓ, L) zero-error list-recoverable if for any sets $L_1, \ldots, L_n \subseteq \Sigma$ with $|L_i| \leq \ell$, the set of codewords $\mathcal{L} = \{c \in C \mid c_i \in L_i \text{ for all } i \in [n]\}$ satisfies $|\mathcal{L}| \leq L$.

You may use the following entropic Brascamp-Lieb inequality without proof. Let A be a finite-dimensional $\mathbb{F}_q$-space, let E be the space of all linear maps from A to $\mathbb{F}_q$, and let $E_1, \ldots, E_n \leq E$. Suppose $\lambda_1, \ldots, \lambda_n \geq 0$ satisfy

$$\dim E - \dim F \leq \sum_{i=1}^{n} \lambda_i \big(\dim E_i - \dim(F \cap E_i)\big) \qquad \text{for every } F \leq E.$$

Then, for every random variable Z on A,

$$H(Z) \leq \sum_{i=1}^{n} \lambda_i H(E_i(Z)).$$

Here H denotes Shannon entropy, and $E_i(Z)$ is the vector obtained by evaluating a basis of E_i at Z.

(a) **Local Properties for zero-error list-recovery.** Let ℓ, L be positive integers. Formulate a family $\mathcal{F}_{\ell,L}$ of local-profiles $(V_1, \ldots, V_n)$ such that an additive code C is (ℓ, L) zero-error list-recoverable if and only if it does not contain such a profile.

(b) **Negative potential from Brascamp-Lieb.** Fix $\alpha \in (0, 1)$. Let ℓ, L be integers such that $L + 1 > \ell^{1/(1-\alpha)}$. Using entropic BL and the potential from Problem 3, prove that for every t-local profile $(V_1, \ldots, V_n)$ you constructed, there is a subspace $W \subsetneq \mathbb{F}_q^t$ such that

$$\Phi_{\mathbb{F}_q^t}(\mathcal{V}, \alpha) - \Phi_W(\mathcal{V}, \alpha) < 0.$$

(c) **Zero-error list recovery of subspace-design codes.** Let C be a τ-subspace-design code, let $\ell \geq 1$ be an integer, fix $0 \leq \alpha < 1$, and put $r = \lceil \ell/(1-\alpha) \rceil$. Deduce from part (a) and (b) that, if $\tau(r) \leq \alpha$, then C is

$$\left(\ell,\ \ell^{1/(1-\alpha)}\right) \text{ zero-error list-recoverable.}$$

Hint: First show that the codewords satisfying all the input lists span a linear space of dimension at most r.

Problem 5: Finding a weakly partition-connected hypergraph

Let $H = (V, E)$ be a hypergraph with $|V| \geq 2$, and let $t \geq 0$. $E = \{e_1, \ldots, e_n\}$ where e_i are subsets of V. We do not require e_i to be non-empty or distinct.

A hypergraph is *t-weakly partition-connected*, or *t-WPC*, if every partition $\mathcal{P}$ of $V = P_0 \sqcup \ldots \sqcup P_m$ satisfies:

$$\sum_{e \in E} \max\big\{|\{j : e \cap P_j \neq \varnothing\}| - 1,\, 0\big\} \geq tm.$$

Suppose

$$\sum_{e \in E} \max\{|e| - 1, 0\} \geq t(|V| - 1).$$

Prove that there is a subset $V' \subseteq V$ with $|V'| \geq 2$ such that

$$H' = (V', E'), \qquad E' = (e \cap V')_{e \in E},$$

is t-WPC.